

Feature Extraction of EEG Signal by Power Spectral Density for Motor Imagery Based BCI

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Abstract— Signals produced from the brain are widely known as Electroencephalogram (EEG) signal interfacing with any communication device creates a unidirectional communicating channel in the absence of neuro-muscular pathways. An effective Brain-Computer Interface (BCI) system basically consists of three operations which are signal recording, feature extraction and classification. Efficient and reliable classification of EEG signal for motor imagery (MI) based BCI system depends on the accuracy of denoising and extracted features of the signal. Extracted features are intended to be lossless key information obtained from a signal that describes a dataset accurately. It is important to minimize the classification complexity and maximize the accuracy. Traditional strategies can be used to process the signal, but the diverseness of the EEG signal conceivably could not be depicted utilizing a linear analytical approach. Hence, this paper adopted the power spectral density (PSD) feature extraction technique to extract the features based on various frequency transformations that enhance the classification performance. Graz BCI competition IV, dataset 2b has been utilized in this paper that consisting of two different classes of motor imagery left-hand and right-hand movement. Overall, 0.61 of Cohen's Kappa accuracy obtained using the LDA classifier.

Keywords— BCI; motor imagery; feature extraction, classification; PSD, LDA

I. INTRODUCTION

Communicating with the people and interacting with machines are the most common task of human beings for daily basis in seek of their own or others benefits. However, this communication or interaction sometimes become impossible for some people due to physical disorder caused by accidents or disease or genetic problem where they lost their normal pathways to sense and express their feelings, thoughts or even to move their limbs. This kind of peoples are called "Locked-in" patients [1] who still can be able to send message and command to the soundings through electroencephalographic (EEG) activity or other electrophysiological measures of brain function that give another non-muscular approach in conveying and communicating with others and this is the process or system called Brain-Computer Interface (BCI).

Studied shows that a neuromuscular patient could learn how to adapt the BCI systems by modifying several features of EEG signal [2]. An effective BCI system basically consists of three particular operations consisting of signal acquisition, feature extraction and classification of the extracted features where feature extraction and classification are the most challenging tasks those define

the accuracy and efficiency of a BCI system. But the classification success and accuracy rely entirely on the accuracy and relevant feature extraction of the EEG signal by minimizing the signal to noise ratio [2]. Therefore, this paper demonstrates the simple but advanced feature extraction technique PSD to maximize the features in the extracted signal through Fast Fourier Transform (FFT).

The paper is organized in five different sections where the materials and methods are described in section two. Following section three gives an overview of the performance measurement with percentage accuracy and in section four results are briefly discussed. Finally, the paper is concluded in section five.

II. RELATED WORKS

An Efficient modality EEG signal is used to record brain signal over a period to diagnose diseases such as Schizophrenia, Epilepsy, Alzheimer and many more. EEG signal has a wide range of usage in many fields including cognitive science, neuroscience and neurolinguistic research, psychophysiological research, and cognitive psychology.

Throughout the years, there are plenty of EEG signal feature extraction and classification technique are being demonstrated on different health applications where the classification accuracy depends on different health domains. In this part, the recent studies on EEG signal pre-processing and classification are briefly explained.

M. R. Hasan et al. proposed Single Layer Perception Learning Algorithm (SLPL) classification method for motor imagery based BCI system. Due to very low computational requirement, they mentioned this technique is suitable for online based BCI. At the beginning, they used PSD feature extraction technique and then classified the signal with SLPL classifier where they achieved 62% Cohen's Kappa accuracy [3]. In another research, they obtained average in training and evaluation both 61% Cohen's Kappa accuracy using PSD feature extraction and LDA classification technique [4].

Ibrahim et al. proposed a robust technique to diagnosis epilepsy where they used discrete wavelet transform (DWT) and Shannon entropy for feature extraction of EEG signal [5]. While nearest neighbour classifier used to classify the signal into normal and epileptic two different categories. The overall performance of the method recorded 97% evaluating with three common measures are specificity, sensitivity, and accuracy.

In another research, EEG signal based in motor imagery BCI, Wavelet Transform (WT) classifier used to control

robotic arm for three arm movements and recorded 91.1% classification performance with a MP-NN [6].

Another approach proposed by Hasan and Bhuiyan for computerized sleep staging method is to detect EEG signal channel based on TQWT technique [7]. Initially this approach crates sub-bands of the recorded signal while sleeping and then the features were extracted using Normal Inverse Gaussian (NIG) method. Later, the extracted features were classified with Adaptive Bosting (AdaBoost) classifier. Finally, the results were evaluated by Cohen's Kappa coefficient, accuracy, sensitivity, and Specificity where Cohen's Kappa coefficient recorded 0.82 (6-class), 0.818 (5-class) and gradually the performance improved for lower classes and its achieved 0.99 (2-class).

From the literature we learned many techniques for feature extraction and classification of EEG signal in different application. However, there is still a gap found for efficient feature extraction and classification technique specially for motor imagery based BCI system for an online and offline EEG application. Therefore, this research has been conducted to achieve better performance.

III. MATERIAL AND METHODS

In BCI design, the brain activities are detected and quantified the features by EEG signal and then these features are translated as a command to utilize the system as user's intent. To accomplish this, three consecutive process called pre-processing that involved with signal denoising, feature extraction and classification are comprised in a BCI system that can be illustrated by the following figure 1.

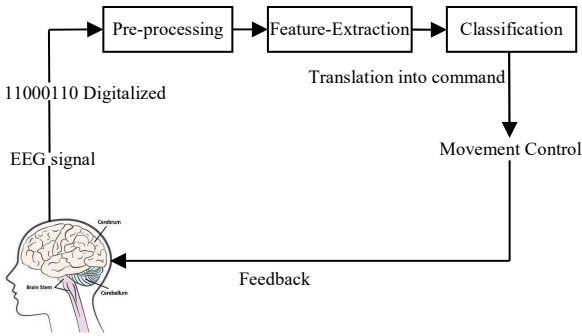


Figure 1: Basic model of a BCI system

Firstly, the artifacts are removed from the raw signal in the pre-processing stage with a bandpass filter. After that, the features are extracted from 2 different channels C3 and C4 by Power Spectral Density feature extraction technique. Two different frequency bands α and β ranging from 8-12Hz and 12-30Hz are extracted from the above stated channels. The data used in this paper collected from BCI competition IV dataset 2b. The data has been recorded and uploaded by Graz University of Technology.

A. Data Selection

BCI competition IV, dataset 2b consisting of total 9 different subjects where each subject performs 5 sessions. First two sessions are without feedback, but last three sessions are with feedback. There are several runs in one session illustrated in following figure 2. The cue-based paradigm is consisted of two classes known as MI based left hand movement (class 1) and right-hand movement (class 2).

The dataset contains 45 general data format files, but 27 files are used as the rest are evaluation purpose only. Each file has EOG (Eye movements) at the start of sessions except B0101T and B0504E. Each file having different number of events and different number of runs. Each run consisting of 20 trails, 10 trails for left hand and rest 10 trails for right hand movement. All trails are randomly distributed, and the length of each trail is different. The sampling frequency is 250Hz and each file is containing signals from different electrodes but C3 & C4 are extracted in this paper.



Figure 2: timing scheme for one session

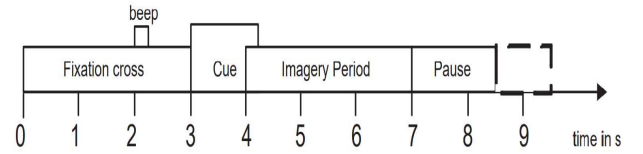


Figure 3: Scenario timeline for the EEG paradigm

Figure 2 represents the timing scheme of one session. Figure 3 illustrates the timing scheme of the paradigm that is as long as 9.5 seconds with a 2.5 second break at last of each session. First 4 seconds time is allocated for the fixation and maintaining the cue and 4-7 seconds dedicated to record the EEG signal of left hand and right-hand movement.

B. Feature extraction

Feature extraction represents key information in a small number of values that extracts from the raw or de-noised EEG signals. Features can be defined as parameters, which provide information about the original structure of a signal [8]. These features should be precise and extracted in such a way thus can be differentiated between other classes to minimize the intra-class feature variances and can achieve high classification accuracy.

Power spectral density (PSD) technique has been manipulated in this research for feature extraction of two class motor imagery EEG signal. Alpha and beta signals are extracted with PSD. The aim of this method is to measure power intensity of the EEG signal. The task has been performed with the help of FFT to obtain frequency domain signal from the filtered time domain signal that manifest frequencies variations [9]. It makes idle to demonstrate power distribution distinguishing between weak and strong frequency spectrums. The PSD is calculated by estimating autocorrelation sequence that is found by nonparametric methods One of these methods is Welch's method [3]. The data sequence $x_i(n)$ can be expressed as:

$$x_i(n) = x(n + iD), \quad n = 0, 1, 2, \dots, M - 1$$

$$\text{while } i = 0, 1, 2, \dots, L - 1; \quad (1)$$

Assuming iD initial state of i th sequence and L represents of data segments. Then the average of these

periodograms express Welch's power spectrum as follows:

$$\hat{P}_{xx}^{(i)}(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} x_i(n)w(n)e^{-j2\pi fn} \right|^2. \quad (2)$$

Here, $w(n)$ represents the window function.

$$P_{xx}^W = \frac{1}{L} \sum_{i=0}^{L-1} \hat{P}_{xx}^{(i)}(f). \quad (3)$$

C. Classification

EEG signals are commonly spoken to in high dimensional element space that is extremely hard to decipher signals physically. Machine Learning strategies are useful for deciphering high dimensional capabilities and break down the attributes of brain signal examples. To perform these tasks there are several classifiers are used like linear and non-linear classifier, neural network, nearest neighbor and sometimes with a combination of two different classifiers [4].

Linear classifiers are used to distinguish classes by discriminant algorithm where the classifier uses linear functions. These are the most mainstream algorithm for BCI applications. Neural network is additionally a broadly utilized classifier in BCI research. Neural system is a gathering of a few artificial neurons which empower to create nonlinear decision limits. Nonlinear classifiers are more suitable compared to discriminative classifiers as they produce nonlinear decision boundaries that empower them to perform more efficiently.

In this paper, Linear discriminant analysis (LDA) classifier is picked to evaluate the PSD feature extraction technique. LDA performs to separate data by similar covariance matrix for different classes. This separation is generally acquired by maximizing distance between class-1 and class-2 but minimizing interclass variance. Three or several hyperplanes can be used in LDA to classify more than two classes that can be defined as

$$J(w) = |m_1 - m_2|^2 / (S_1^2 + S_2^2) \quad (4)$$

The general concept here is that LDA creates a new axis between two classes and projects the data onto that new axis in a way that maximizes the separation of two classes. In LDA, the separation is called scatter represented by S -squared and m is the mean value for each class [10].

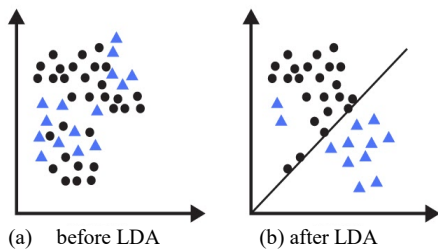


Figure 4: General concept of LDA classification

Although LDA provides better performance for linear classification but it is limited in Small Sample Size data reduction and nonlinear data separation.

D. Performance Measurement

The performance of the proposed feature extraction technique for the BCI has been evaluated by class recognition accuracy upon LDA classification. It has been done by measuring the left-hand and right-hand movement detection accuracy. Besides that, Cohen's kappa coefficient is used to assess the performance accuracy.

a. Accuracy

The accuracy for each instant of Motor imagery 2b dataset has been computed by comparing the classifier's output (estimated label = the sign of $d(m)$, where right is positive, and left is negative) with the actual left or right trials to prepare a confusion matrix (CM) [3]. Then the left and right-hand movement accuracy are calculated separately in percentage by the following formulae:

$$\text{Left accuracy} = \frac{\text{true negative in CM}}{\text{number of actual left trials}} \times 100 \quad (5)$$

$$\text{Right accuracy} = \frac{\text{true positive in CM}}{\text{number of actual right trials}} \times 100$$

b. Cohen's Kappa Coefficient

Cohen's Kappa coefficient known as Cohen's Kappa is a statistical measure of inter-rater agreement of the agreement between two raters for categorical items. It computes the percentage accuracy for each instant for the actual trials and estimated trials observed as classifier's output. From the definition, Cohen's Kappa can be written as [11]

$$\kappa = \frac{P_o - P_c}{1 - P_c} \quad (6)$$

Here, the relative observed agreement between two raters is represented by P_o , and P_c is the hypothetical probability of chance agreement. The maximum Kappa value is 1 when both raters are in positive agreement. If there is no agreement between the raters, then $\kappa=0$.

IV. RESULTS AND DISCUSSION

The research gives a better output using PSD feature extraction technique-based classification for EEG signal. The results are obtained from 9 different subjects and calculated their average peak value. The following results in table 1 shows the overall accuracy in training stage, on the other hand table 2 shows in evaluation stage. The average overall max Kappa value is calculated in Table 3. (From the results depicted in the tables below show the better overall accuracy and max Kappa value compared to the result obtained from BCI competition IV).

The average maximum accuracy for both training and evaluation stage is 0.61 max Kappa (Table 3) achieved where 0.57 computed in BCI competition IV. Therefore, PSD is an optimized feature extraction technique for MI based BCI system

TABLE 1: CLASSIFICATION RESULT BASED ON PSD FEATURE EXTRACTION FOR EVALUATION STAGE

Subject	Training stage	
	Overall accuracy Max (in %)	Max. Kappa
B01	62	0.78
B02	54	0.67
B03	93	0.93
B04	78	0.91
B05	85	0.68
B06	70	0.63
B07	89	0.42
B08	97	0.65
B09	85	0.89
Average	79	0.73

TABLE 2: CLASSIFICATION RESULT BASED ON PSD FEATURE EXTRACTION FOR TRAINING STAGE

Subject	Evaluation stage	
	Overall accuracy Max (in %)	Max. Kappa
B01	59	0.26
B02	64	0.54
B03	91	0.87
B04	67	0.31
B05	77	0.57
B06	72	0.48
B07	89	0.49
B08	91	0.69
B09	58	0.43
Average	74	0.52

TABLE 3: AVERAGE CLASSIFICATION RESULT FOR BOTH TRAINING AND EVALUATION STAGES

Both training and evaluation stages	
Subject	Average max Kappa
B01	0.47
B02	0.48
B03	0.92
B04	0.60
B05	0.61
B06	0.50
B07	0.53
B08	0.72
B09	0.68
Average	0.61

V. CONCLUSION

This paper explained and implemented PSD feature extraction method that shows remarkably better and very consistent of EEG signal classification over LDA classifier for Motor imagery based BCI system. It also aimed to achieve the optimized frequency ranging from 8-30Hz by removing the artifacts and denoising the signal with

bandpass filter for better feature extraction and classification purpose. This research obtained 0.61 max Kappa accuracy that is higher than the (BCI competition IV 57% Kappa coefficient). Besides that, Simple computational technique makes PSD suitable and reliable for EEG signal feature extraction. It is recommended in future to work on the topic to identify a robust feature extraction technique utilizing metaheuristics-based algorithms that would give better classification accuracy.

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